

Subroutine	Description	Inputs	Outputs
GET_INT_VARS	Reads in NetCDF file (boxm_ds.nc), containing meteorology, background chemistry and emissions data for each time step. FORTTRAN variables allocated and output to the program. At the start time of the simulation, Y is the sum of BG_CHEM and EMI. At subsequent time steps, Y is inferred from the previous time step (YP).	MET: GPAT Meteorology Dataset BG_CHEM: GPAT Background Chemistry Dataset EMI: Emissions Dataset	TEMP(CELL): Air temperature M(CELL): Number density of air H2O(CELL): Water vapour concentrations O2(CELL): Oxygen concentrations N2(CELL): Nitrogen concentrations SZA(CELL): Solar Zenith Angle BG_CHEM(CELL,SPECIES): Background chemical concentrations EMI(CELL,EMI_SPECIES): Emissions concentrations Y(CELL,SPECIES): Total concentrations YP(CELL,SPECIES): Previous total concentrations
CALC_AEROSOL	Calculates aerosol mass of BGOAM to be 0.7 (user defined). SOA is calculated to be the sum of the products of SOA species and their corresponding rate coefficients. MOM is the sum of BGOAM and SOA.	Y(CELL,SPECIES): Total concentrations	BGOAM(CELL): Background organic aerosol mass SOA(CELL): Secondary organic aerosol MOM(CELL): Mass of organic matter
CHEMCO	This firstly calculates all simple and complex rate coefficients. Then it calculates all 512 thermal rate coefficients which can vary with TEMP, M and BR.	TEMP(CELL): Air temperature M(CELL): Number density of air BR: Branching ratio	RC(CELL,THERM_COEFF): Thermal rate coefficients
CALC_J	This subroutine calculates 34 photolysis parameters from SZA.	SZA(CELL): Solar Zenith Angle	J(CELL,PHOTOL_PARAM): Photolysis parameters
PHOTOL	This is the photolysis subroutine in which 96 photolysis rate coefficients are calculated from J and BR.	J(CELL,PHOTOL_PARAM): Photolysis parameters BR: Branching ratio	DJ(CELL,PHOTOL_COEFF): Photolysis rate coefficients